ECOTITANS

FOOD WASTE REDUCTION IN HOSTEL

Project by-

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PROBLEM STATEMENT-

In most hostels, food is prepared based on rough estimates rather than actual demand. This leads to:

- **Excess food preparation**
- **High food wastage**
- **Inefficient use of resources**
- **Increased operational costs**
- **Environmental impact due to wasted food**

Therefore, there is a need for a digital system that helps predict meal attendance and optimize food preparation.

OBJECTIVES:-

- To predict the number of students attending meals.
- To help kitchen staff prepare food according to demand.
- To track leftover food and analyze waste patterns.
- To encourage responsible food consumption among students.

PROPOSED SOLUTION:-

A smart food demand prediction system that includes a mobile application or website where students can confirm their meal attendance in advance.

The system will collect attendance data and use it along with other factors such as day of the week, holidays, and historical patterns to predict the expected number of students for each meal.

The predicted attendance will be displayed on a dashboard for the kitchen staff, helping them prepare the appropriate amount of food.

SYSTEM FEATURES:-

1. MEAL ATTENDANCE PREDICTION

Students can mark their availability for upcoming meals (breakfast, lunch, dinner) through the app or website

Features:-

- **Simple yes/no confirmation**
- **Easy interface for students**

IMPACTS:

- Helps estimate the number of students attending meals
- Reduces uncertainty for kitchen staff

2. AI-BASED DEMAND PREDICTION

The system will analyze:

- Historical attendance data
- Day of the week
- Holidays
- Exam periods
- Special events

Using this data, the system will generate a prediction of the expected number of students for each meal.

3. KITCHEN DASHBOARD

A dedicated dashboard will be available for mess staff.
The dashboard will show:-

- **Confirmed meal attendance**
- **Historical data**

IMPACTS:

- Better planning for food preparation
- Efficient resource management

4. FOOD WASTE TRACKING

After each meal, the mess staff will enter the amount of leftover food.
The system will store:

- Yes or No for breakfast , lunch and dinner

IMPACTS:

- Enables improvement in future predictions

5. ANALYTICS AND REPORTS

The system will generate reports such as

- Attendance trends
- Breakfast, lunch and dinner
- Waste reduction progress

These reports will help hostel management evaluate system effectiveness.

IMPACTS:

- Improves participation
- Makes predictions more accurate

6. SYSTEM ARCHITECTURE

Users of the System

1. Students

- Confirm meal attendance
- Receive reminders

2. Mess Staff

- View expected number of students
- Record leftover food

3. Admin/Hostel Management

- View reports and analytics
- Monitor waste reduction

TECHNOLOGY STACK UTILIZED-

BACKEND DEVELOPMENT:
Developed using Python to handle application logic, data processing, and system operations.

PLATFORM:
Designed a Google form and console based program.

DEVELOPMENT TOOLS:
Tools such as Anaconda ,Jupiter

Conclusion -

The proposed Smart Food Demand Prediction and Management System aims to address the issue of food wastage in hostels through a combination of student participation, data analysis, and digital tools. By predicting meal demand and monitoring food waste, the system can help create a more efficient and sustainable hostel mess environment